Chapter 3

Research and Development Methodology

This study utilized the Research and Development (R&D) methodology to guide the systematic creation and evaluation of the proposed system. The R&D approach is appropriate for this research because it focuses on developing a functional software product while ensuring that it effectively addresses the identified institutional problems. Through this methodology, the researchers were able to transform the gathered requirements into a structured system design and working application. It also allowed continuous refinement of system components through iterative testing and validation. This process ensured that both user requirements and institutional standards were consistently incorporated throughout development.

The study integrates research-driven analysis and software engineering principles to guide the creation of a scalable, secure, and efficient web-based system. It describes the selected Software Development Life Cycle (SDLC) model, development phases, system validation procedures, feasibility analysis, system architecture design, database modeling, security mechanisms, and implementation environment. Through this structured framework, the methodology ensures that the proposed system meets functional, technical, operational, and quality standards.

Software Development Life Cycle Model

This study adopted the Rapid Application Development (RAD) model as the Software Development Life Cycle (SDLC) approach in the design, development, and implementation of the proposed Integrated Student Organization Management System for the National Aviation Academy of the Philippines – Villamor. The SDLC provides a structured framework for developing information systems through defined phases, ensuring systematic requirements analysis, solution design, system development, testing, and deployment while maintaining quality and operational efficiency.

The RAD model was chosen due to its focus on quick prototyping, iterative development, and ongoing user participation, which are crucial in meeting the changing and evolving needs of managing student organizations. Considering that the current procedures at NAAP–Villamor are manual and disjointed, regular input from the Office of Student Affairs, organization leaders, and students is essential to ensure that the system accurately mirrors actual workflows. RAD provides adaptability in enhancing system capabilities throughout development, shortens overall development duration, and guarantees prompt identification and resolution of problem

The RAD-based SDLC implemented in this study consisted of four major phases: Requirements Planning, User Design, Construction, and Cutover. Each

phase was systematically executed to ensure the successful development, validation, and deployment of the proposed system.

Figure 1 illustrates the Rapid Application Development (RAD) model adopted in this study, demonstrating the iterative and incremental progression of system development phases. The model emphasizes rapid prototyping, active user participation, and continuous stakeholder feedback to refine system requirements and functionalities. Through repeated design, development, and evaluation cycles, the system was progressively improved until it met the intended objectives and user expectations.

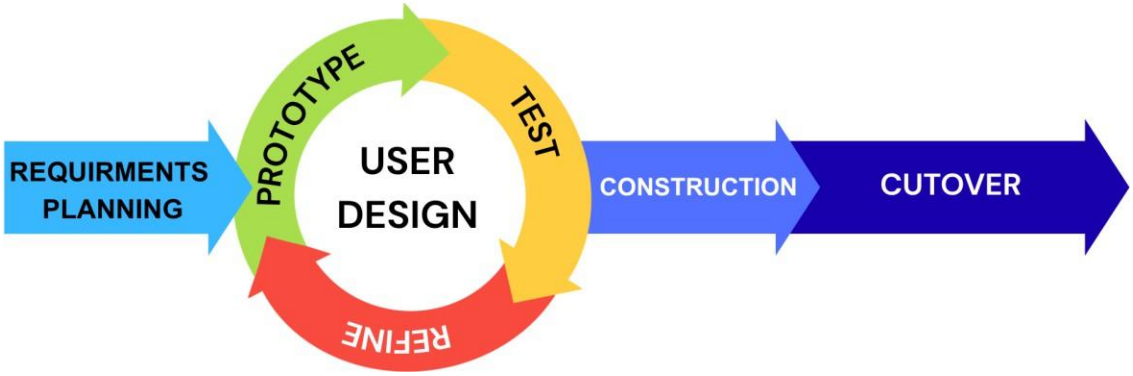

Figure 1. Rapid Application Development

Requirements Planning Phase

During the Requirements Planning phase, the researchers identified and analyzed the problems, objectives, and system requirements of the proposed Integrated Student Organization Management System. An assessment of the existing student organization management procedures was conducted involving the Office of Student Affairs, student organization officers, and students.

This study observed that the current system involved manual submission of documents, physical record keeping, and the use of multiple communication platforms. These practices resulted in delays in approval processes, difficulty in tracking documents, inefficient record management, and limited accessibility of organizational information.

To address these issues, the researchers gathered and defined both functional and technical requirements necessary for system development. The identified system requirements included. Secure login and authentication system, role-based access for students, organization officers, and administrators, digital submission and approval of organizational documents, Organization and membership management module, Notification and communication system, Dashboard and reporting features for monitoring organizational activities.

User Design Phase

In the User Design phase, the researchers designed the system

architecture, user interface, and system workflows based on the identified requirements. System prototypes were developed to visualize the layout, features, and overall functionality of the proposed system.

Prototypes of key system components included, login and authentication interface, user dashboards for students, organization officers, and administrators, document submission and approval interface, organization management module, notification and communication features.

Construction Phase

During the Construction phase, the researchers developed the actual web-based Integrated Student Organization Management System based on the approved design and requirements. The system design was translated into a functional application using appropriate programming tools, development frameworks, and database management systems.

The implemented system directly addressed the issues identified in the existing manual and fragmented processes by incorporating key components such as a secure user authentication system, role-based access control, digital document submission and approval, organization and membership management, notification and communication features, and a reporting and analytics dashboard. These components were specifically designed to streamline workflows, improve data accessibility, and enhance administrative efficiency within the Office of

Student Affairs and student organizations.

Throughout this phase, continuous system testing was conducted to ensure proper functionality, usability, security, reliability, and efficiency. Identified errors were corrected, and performance improvements were applied to ensure that the system met established quality standards.

Cutover Phase

The Cutover phase marked the final stage of the RAD-based SDLC, where the system was tested, deployed, and implemented for actual use. Final system testing ensured all features functioned properly, while User Acceptance Testing (UAT) was conducted with selected students, organization officers, and administrators to evaluate usability, performance, and requirement compliance. System orientation and demonstrations were also provided to familiarize users with its features and operations. After validation and approval, the Integrated Student Organization Management System was officially deployed.

This phase ensured successful implementation and provided users with a secure, efficient, and accessible platform for managing student organization activities. Through the RAD methodology, the researchers developed a reliable, user-centered system that meets institutional requirements, improves operational efficiency, enhances accessibility, and offers a scalable and centralized solution for student organization management.

Development Methodology Justification

The Rapid Application Development (RAD) model was selected as the Software Development Life Cycle (SDLC) approach because it provides a flexible, efficient, and user-centered framework that effectively addresses the limitations of the institution's previous manual processes. The former system relied heavily on physical documentation, fragmented communication, and manual record-keeping, which resulted in delays, inefficiencies, and difficulty in monitoring organizational records and transactions. The RAD model emphasizes rapid prototyping, iterative refinement, and continuous user involvement, allowing the researchers to develop system prototypes and gather feedback from users throughout the development process. This ensured that the system's features and functionalities were aligned with the actual needs of the Office of Student Affairs, student organization officers, and members.

The RAD model enabled faster system development and deployment within the given academic timeframe without compromising system quality, security, and performance. Its iterative approach allowed the researchers to identify and resolve issues early, improve usability, and enhance system reliability through continuous testing and refinement. This approach also promoted active user participation, which strengthened user validation and increased the likelihood of successful system adoption.

Furthermore, the RAD model is closely aligned with Agile principles, as both methodologies emphasize iterative development, flexibility, and continuous user feedback. Similar to Agile, RAD supports incremental delivery of system components, adaptive planning, and frequent validation with stakeholders. This alignment allows the development process to remain responsive to changing requirements while maintaining a strong focus on user needs and system functionality. By integrating Agile-like practices within the RAD framework, the study ensures a more adaptive, user-driven, and efficient development process. The RAD-based SDLC was considered the most appropriate methodology for ensuring efficient development, effective user validation, and successful implementation of the proposed web-based Integrated Student Organization Management System.

System Development Framework

The System Development Framework outlines the structured methodology used in developing the proposed Integrated Student Organization Management System for the National Aviation Academy of the Philippines – Villamor. The framework is based on the Rapid Application Development (RAD) model under the Software Development Life Cycle (SDLC), guiding the system through requirements analysis, design, development, testing, and deployment. It ensures a systematic, efficient, and user-centered development process aligned with

institutional requirements.

Analysis of the Existing System

The Analysis of the Existing System provides a comprehensive examination of the institution's current evaluation procedures, focusing on how faculty and student assessments are conducted, the tools and processes involved, and the stakeholders participating in the workflow. This section identifies existing limitations and operational challenges such as delays in data processing, manual errors, limited accessibility, fragmented records, and concerns related to data security and reporting accuracy. Through problem analysis, the root causes of inefficiencies are determined, followed by a requirements gap analysis that compares the present system's capabilities with the functional and non-functional requirements of an effective, secure, and CHED-compliant evaluation platform. The section also explores possible alternative solutions, evaluates their feasibility in terms of technical, operational, economic, and schedule considerations, and discusses proposed technology alternatives that may enhance automation, reliability, scalability, and overall system performance, thereby establishing the foundation for the development of an improved and fully integrated evaluation system.

Current Evaluation Procedures

The management processes of student organizations at the National Aviation Academy of the Philippines – Villamor are presently marked by manual, decentralized, and disjointed methods overseen by the Office of Student Affairs (OSA). The majority of administrative duties depend significantly on paper documents and in-person collaboration. This conventional method restricts effectiveness and hinders overall processes.

Student organization officers must provide financial reports, accomplishment reports, event proposals, and other relevant documents in physical copy. OSA staff manually examines these submissions to confirm adherence to institutional policies. Consequently, approvals frequently necessitate several in-person visits before final authorization is achieved.

Interaction between student groups and the OSA occurs via distinct and fragmented communication channels. Announcements are spread unevenly, resulting in disparities in information distribution. This disjointed communication framework hinders coordination effectiveness among participants.

Documents are kept either in tangible filing systems or in separate digital files. Due to this configuration, monitoring in real-time and centralized oversight of organizational activities cannot occur. Accessing old documents and confirming records can thus turn into a lengthy and ineffective process.

Although institutional policies governing student organizations are already established, there is no centralized information system to operationalize them. The absence of a structured digital workflow prevents seamless alignment between governance requirements and administrative execution. Consequently, a gap exists between policy implementation and actual operational practices within the institution.

Limitations and Problem Analysis

The reliance on manual and fragmented processes results in several operational limitations within the institution. Document submissions and approvals are frequently delayed due to physical routing and in-person coordination requirements. These delays negatively affect the overall efficiency of student organization management.

In addition to increasing the administrative load on OSA employees and student group leaders, this procedure decreases accountability and transparency. Accurately tracking submission and approval statuses is made more difficult by manual tracking. Coordination is therefore more time-consuming and prone to human error.

The lack of a centralized digital archive results in uneven record-keeping methods. Accessing past reports and confirming historical information becomes difficult and ineffective. This scenario diminishes the precision and dependability

of tracking activities.

The OSA's ability to oversee several organizations at once is limited by the absence of real-time monitoring tools. Lacking a cohesive system makes it challenging to execute proactive monitoring and prompt actions. These restrictions undermine efficient administration and organized institutional oversight.

Students face diminished access to services and updates related to the organization. The absence of a centralized platform for announcements and information sharing leads to communication gaps. As a result, student involvement and participation in organizational events may be negatively influenced.

Requirements Gap Analysis

The existing operational landscape shows a notable gap in policy execution within the institution. Although national policies promoting digital transformation and effective governance have been established, existing processes still predominantly depend on manual approaches. This situation highlights a gap between regulatory systems and real institutional operations. To bridge this gap, the suggested web-based Integrated Student Organization Management System aims to implement these policies by digitizing procedures, standardizing workflows, and facilitating efficient, transparent, and policy-compliant administration of student organizations within the institution.

National directives like CHED Memorandum Order No. 09 promote Smart Campus initiatives and the incorporation of technology in educational institutions. In the same way, Republic Act No. 11032 requires streamlined, digital, and effective public service provision. Furthermore, ARTA Memorandum Circular No. 2020-03A strengthens mechanisms for zero-contact and digital transactions in processes related to government.

The present operation reveals a notable gap in policy implementation within the organization. Despite having national policies that advocate for digital transformation and efficient governance, current processes are mainly manual. This scenario underscores a gap between regulatory systems and real institutional behaviors.

National policies like CHED Memorandum Order No. 09 support Smart Campus programs and the incorporation of technology in educational institutions. In the same way, Republic Act No. 11032 requires streamlined, digital, and effective public service provision. Furthermore, ARTA Memorandum Circular No. 2020-03A strengthens the implementation of zero-contact and digital transaction methods in government procedures.

Alternative Solutions and Feasibility Analysis

When assessing possible system options, keeping the current manual system will involve no extra development expenses; nonetheless, it will continue

the current operational inefficiencies, processing lags, and disjointed coordination methods. Additionally, this method would lead to only limited adherence to national policies advocating for digital governance and efficient public service delivery.

A different approach includes using generic or subscription-driven third-party platforms. Although these systems provide ready-made functionalities, many rely on subscriptions, are costly, or have restricted capabilities in free-tier versions. Moreover, numerous existing platforms do not entirely correspond with the governance, regulatory, and operational needs of higher education institutions within the Philippine setting.

Additionally, generic systems have restrictions related to customization, scalability, advanced security features, and the integration of policies. Their standardized architectures might not sufficiently accommodate workflows, approval hierarchies, and compliance needs specific to the institution. Consequently, dependence on outside platforms could create operational limitations and have lasting financial effects.

Considering these factors, the creation of a secure, centralized, and institution-specific integrated management system is recognized as the most feasible solution. The suggested online platform for NAAP–Villamor aims to facilitate electronic document submission, organized approval processes, centralized data oversight, and real-time administrative tracking. Furthermore, the system guarantees adherence to regulatory guidelines, such as CHED

Memorandum Order No. 09, Republic Act No. 11032, and ARTA Memorandum Circular No. 2020-03A, thus improving institutional effectiveness, transparency, and compliance

Proposed Technology Alternatives

The suggested technological option entails creating a secure, centralized, and institution-tailored web-based Integrated Management System for student organizations at NAAP–Villamor. This system aims to transform disjointed and manual processes into a cohesive digital framework. By means of system integration, processes within the organization will be made more efficient and uniform.

The system will enable electronic document submission, organized approval processes, centralized digital record keeping, and integrated communication channels. By integrating organization-related transactions into one platform, redundant processes will be removed and the need for physical coordination will be reduced. This integration will allow for real-time observation and improved management of submissions and approvals.

The suggested platform will greatly improve administrative supervision by the Office of Student Affairs. Automated processes and unified dashboards will enable proactive oversight of several organizations at the same time. Consequently, transparency, accountability, and operational oversight will be

enhanced.

The system will enhance access to information and services for students and organizational officials. Notifications, progress reports, and necessary paperwork will be accessible through a unified secure platform. This centralized access will enhance communication and boost student involvement.

The suggested system is in accordance with national digital governance structures, such as CHED Memorandum Order No. 09, Republic Act No. 11032, and ARTA Memorandum Circular No. 2020-03A. By facilitating efficient, electronic, and no-contact transactions, the platform will implement governance needs into organized digital workflows. This infrastructure aligned with policy will improve efficiency, compliance, transparency, and overall effectiveness in managing student organizations.

Feasibility Study

The feasibility study assesses the practicality of creating and executing the proposed Integrated Student Organization Management System at the National Aviation Academy of the Philippines – Villamor. This evaluation determines whether the project can be effectively implemented within the institution's technological resources, operational environment, financial capacity, and projected timeline. The study focuses on four key dimensions: Technical Feasibility, which evaluates the availability of required technologies, infrastructure, and technical expertise; Operational Feasibility, which examines the system's

alignment with existing workflows, user readiness, and institutional policies; Economic Feasibility, which analyzes the cost requirements and potential long-term benefits of implementation; and Schedule Feasibility, which assesses whether the system can be developed and deployed within the proposed timeframe.

To further support the feasibility of the proposed system, recent project studies and related systems (2022–2026) were considered, particularly those involving web-based management platforms and student information systems. Contemporary studies indicate that centralized and web-based systems incorporating secure authentication, role-based access control, digital workflows, and real-time analytics significantly enhance administrative efficiency, data accuracy, and transparency in academic institutions [64][65]. Furthermore, recent research highlights that integrated dashboards, automated notification systems, and centralized databases improve communication, coordination, and decision-making among institutional stakeholders [66][67]. These findings support the inclusion of similar functionalities in the proposed system, confirming its technical viability and alignment with current trends in digital transformation within higher education.

Technical Feasibility

The technical feasibility of the proposed Integrated Management System Enhancing Student Organization Member Accessibility and Engagement at the National Aviation Academy of the Philippines – Villamor is established through the availability of required technologies, infrastructure, and technical competencies necessary for system development and deployment. The proposed system is designed as an online platform, guaranteeing access via common web browsers without the need for specialized hardware or proprietary software. As the institution currently employs internet access and computer labs for both academic and administrative functions, the current infrastructure is adequate for system implementation. The architecture of the platform is based on a client-server model, where users (students, officers of student organizations, and administrators from the Office of Student Affairs) interact with the system through web browsers, and centralized servers manage data processing, authentication, storage, and workflow automation.

From a development standpoint, the system was implemented using widely adopted web technologies to ensure reliability, maintainability, and scalability. The front-end interface was developed using HTML, CSS, and JavaScript, which provided the structural framework, visual design, and interactive functionalities of the web application. On the server side, PHP was utilized as the primary back-end scripting language to manage core system operations, including form processing,

business logic implementation, session handling, and overall application control. The system follows a hybrid monolithic architecture, where all core modules are integrated within a single application while maintaining logical separation to improve organization and maintainability.

For data management, SQL was used within a relational database management system to store, retrieve, and manage essential records such as business documents, user accounts, approval transactions, and system logs. PHP was responsible for securely connecting the application to the database to ensure efficient data handling and integrity. During development, XAMPP served as the local hosting environment, providing Apache, MySQL, and PHP in an integrated setup for testing and refinement. For deployment, the system was migrated to a cloud-based hosting platform to enhance accessibility, scalability, and availability. The system can operate on standard personal computers (PCs) or laptops with modern web browsers and stable internet connectivity, making it accessible to students, organization officers, and administrators without requiring specialized hardware.

The system incorporates secure login mechanisms, role-based access control (RBAC), encrypted data transmission (HTTPS), and centralized database management to protect sensitive student and organizational data. Security design considerations align with ISO/IEC 25010 quality characteristics, particularly in terms of security, reliability, performance efficiency, and maintainability. The

development approach follows a structured Software Development Life Cycle (SDLC), specifically the Rapid Application Development (RAD) model, which emphasizes iterative prototyping, continuous testing, and user feedback integration. This ensures that technical issues can be identified and resolved early during development, reducing implementation risks.

In terms of deployment, the platform may be hosted on a local institutional server or a secure cloud-based hosting environment, depending on available institutional resources. Both deployment strategies are technically viable, as the system does not require high computational complexity or advanced hardware specifications. Scalability issues can be managed by optimizing servers and implementing database indexing if user access grows over time. Considering the presence of web technologies like HTML, CSS, JavaScript, PHP, and SQL, along with the existing internet infrastructure of the institution, standard hardware capabilities, and the technical skills of Information Systems students and faculty, the suggested system is technically viable, offering accessible software resources, controllable implementation challenges, and an institutional environment that can support sustained deployment and operation.

Operational Feasibility

The operational feasibility of the proposed Integrated Management System for Improving Student Organization Member Accessibility and Engagement at the National Aviation Academy of the Philippines – Villamor is based on the

institution's current administrative framework, user preparedness, and coherence with existing governance regulations and operational methods. Currently, the management of student organizations at NAAP–Villamor is carried out through manual, decentralized, and disjointed processes under the oversight of the Office of Student Affairs (OSA). Submission of documents, approvals, oversight, and communications are conducted through tangible paperwork and distinct communication methods, leading to delays, reduced transparency, and heightened administrative burden. The suggested system does not present completely new procedures; instead, it digitizes and enhances current workflows like financial report submissions, accomplishment reporting, event proposals, monitoring, and approval routing.

The system is designed to support three primary user groups: students, student organization officers, and OSA administrators. Role-based dashboards ensure that each user category interacts only with features relevant to their responsibilities, minimizing complexity and reducing training requirements. Students gain centralized access to organization information, services, and event participation records, while organization officers manage submissions and monitoring tasks. OSA administrators retain oversight authority through verification and approval functionalities. This alignment between system roles and existing institutional roles enhances user acceptance and operational compatibility.

Operational feasibility is further strengthened by the system's compliance with Republic Act No. 11032 and ARTA Memorandum Circular No. 2020-03A, which promote zero-contact transactions and streamlined service delivery. By automating document submission, approval workflows, and monitoring, the system operationalizes these mandates within the student organization governance framework. Additionally, alignment with CHED Memorandum Order No. 09, s. 2020 (Smart Campus Initiative) ensures that the system supports institutional digital transformation goals.

In terms of user readiness, the primary stakeholder students and administrators are already accustomed to using web-based platforms for academic and communication purposes. The proposed system's web-based interface, notification features, and centralized dashboards are consistent with commonly used digital systems, reducing resistance to adoption. Training requirements are minimal and can be addressed through orientation sessions, user manuals, and guided demonstrations during implementation. Although limitations exist, such as dependency on stable internet connectivity and phased deployment among selected organizations, these constraints do not prevent system operation. Instead, they represent manageable implementation considerations that can be addressed through institutional IT support and gradual rollout strategies.

From an economic perspective, studies on web-based information systems indicate that transitioning from manual to digital processes significantly reduces

long-term operational costs associated with paper-based documentation, physical storage, and repetitive administrative labor [68][69]. Initial development costs, including system design, programming, and deployment, are offset by long-term benefits such as improved process efficiency, reduced processing time, and minimized resource consumption. Additionally, the use of open-source technologies and scalable web-based architectures further lowers implementation and maintenance costs while ensuring system sustainability.

The proposed system is operationally feasible because it aligns with existing institutional workflows, supports established governance structures, complies with national policies on digital service delivery, and matches the digital literacy level of its intended users. The system enhances efficiency and oversight without requiring major organizational restructuring, ensuring practical implementation within NAAP–Villamor.

Economic Feasibility

The proposed Integrated Management System Enhancing Student Organization Members Accessibility and Engagement in the National Aviation Academy of the Philippines – Villamor is economically feasible because its expected benefits outweigh the costs of development, implementation, and maintenance. The study aims to improve governance efficiency, accessibility, records consolidation, digital workflows, and administrative monitoring through a

centralized web-based platform. In addition, the system is expected to address the limitations of the current manual and fragmented processes by providing a more organized, accessible, and reliable approach to student organization management. The system is also designed to be scalable, allowing it to accommodate a growing number of users, organizations, and records without requiring significant additional costs in the future.

The main costs of the proposed system involve software development, database setup, testing, deployment, and user orientation. Since the system is developed using the Rapid Application Development (RAD) model, resources are used more efficiently through rapid prototyping, iterative refinement, and continuous user feedback. This development approach helps minimize unnecessary expenses by ensuring that system features are refined early and aligned with actual user needs. Furthermore, the system is designed with a user-friendly interface, reducing the need for extensive training and lowering implementation costs. Its capability for future integration with other institutional systems also increases its long-term value and sustainability.

The current manual process at NAAP–Villamor also creates operational costs in terms of time, effort, and administrative workload. Physical submissions, fragmented communication, manual record-keeping, and difficulty in tracking documents contribute to delays, inefficiencies, and inconsistent monitoring of organizational activities. These issues may also affect the timely processing of

requests and the accessibility of important information for both students and administrators. By digitizing submission, approval, notification, and monitoring processes, the proposed system automates repetitive administrative tasks, reduces human error, and improves overall coordination. The centralized data management feature ensures that all records are stored in a single, reliable platform, minimizing data redundancy and loss while improving consistency and accessibility.

The system also offers tangible and intangible benefits. Tangible benefits include reduced paper use, faster processing, improved record management, and more efficient handling of student organization documents and requests. Intangible benefits include enhanced accessibility through a web-based platform, improved transparency and accountability through monitoring features, and stronger data security through role-based access control. In addition, the system provides real-time access to information, enabling users to stay updated and make timely decisions. Its built-in data analytics and reporting capabilities further support administrators in evaluating organizational performance and making informed decisions.

The proposed system is economically feasible because it provides a practical and sustainable solution to the current manual and fragmented student organization management process. Although initial development requires investment, the long-term gains in efficiency, reduced operational costs, improved

data reliability, and enhanced administrative effectiveness make the system cost-justified for NAAP–Villamor. The inclusion of backup mechanisms ensures data reliability and continuity of operations, while its alignment with digital transformation goals strengthens its relevance and long-term impact. Overall, the expected improvements in service delivery, transparency, and institutional resource management further support the value of implementing the proposed system in the institution.

Schedule Feasibility

The schedule feasibility of the proposed Integrated Management System Enhancing Student Organization Members Accessibility and Engagement in the National Aviation Academy of the Philippines – Villamor was evaluated by assessing the development timeline, system testing schedule, and deployment plan in relation to the operational and academic activities of the institution. The assessment aimed to determine whether the system could be designed, developed, tested, and implemented within a timeframe that supports institutional requirements without disrupting student organization processes and administrative functions. The proposed system is intended to improve accessibility, document handling, monitoring, and student engagement through a centralized web-based platform.

The system development follows the Rapid Application Development (RAD) model, which supports fast and iterative development through continuous prototyping, user feedback, testing, and refinement. The SDLC adopted in the study consists of four major phases, namely Requirements Planning, User Design, Construction, and Cutover. Each phase was systematically carried out to ensure that system requirements were identified properly, prototypes were evaluated by intended users, and the final system was tested and deployed within the academic timeframe. The iterative nature of the RAD model also helps reduce the risk of delays by allowing issues to be identified and addressed at an early stage of development.

The implementation schedule was designed to ensure that the system could be deployed before full institutional use so that students, student organization officers, and Office of Student Affairs administrators would have sufficient time for familiarization, validation, and adjustment. This timeline allows adequate time for system testing, User Acceptance Testing (UAT), system orientation, and final improvements prior to actual implementation. Conducting deployment after the completion of development and validation activities helps ensure that the system functions properly and minimizes disruption to document submissions, approval workflows, notifications, and monitoring processes.

Potential scheduling risks include technical issues during development, limited availability of stakeholders during consultation and prototype evaluation,

revisions resulting from user feedback, and possible delays in testing and deployment. These risks are mitigated through the RAD approach by supporting continuous stakeholder involvement, iterative refinement, and flexible adjustment of system components throughout the development cycle. The use of a web-based platform also contributes to schedule feasibility because it simplifies deployment and allows users to access system features without complex installation requirements.

Additionally, user orientation and system demonstrations were incorporated into the implementation timeline to support readiness and improve adoption among intended users. This is important because the system involves multiple user groups with different roles and functions, including students, organization officers, and administrators. Since the proposed platform includes role-based access, digital document submission and approval, notification services, and reporting features, providing sufficient time for orientation and validation supports smoother implementation and more effective transition from the previous manual process.

Based on the alignment of the development phases with institutional requirements, the inclusion of testing and orientation activities, and the flexibility of the RAD-based development framework, the proposed system is considered schedule feasible. The structured yet iterative nature of the development process supports timely completion and implementation of the system without significantly disrupting the student organization management operations of NAAP–Villamor.

Cost Benefit Analysis

Overview

This cost-benefit analysis evaluates the economic feasibility of developing and implementing the Integrated Management System Enhancing Student Organization Members' Accessibility and Engagement in the National Aviation Academy of the Philippines – Villamor. The analysis compares the estimated development and operational costs of the proposed system with its expected tangible and intangible benefits over a three-year period, which is a common evaluation horizon for academic system feasibility studies.

Cost Analysis

Development Cost

The development cost covers the planning, design, module development, testing, deployment, and initial system setup of the proposed web-based integrated management system.

Table 1. Development Cost

Cost Component	Description	Estimated Cost (PHP)
Requirements Analysis and Sprint Planning	Stakeholder consultation, process mapping, backlog creation, and planning of system iterations	35,000
System Design and Architecture	User interface design, database design, system structure, and module planning	45,000
Core Module	Development of user management, organization management,	150,000

Development	membership records, announcements, events, and document-related modules	
Dashboard and Reports Development	Creation of monitoring tools, summaries, activity reports, and engagement-related dashboards	30,000
Testing and Iteration	Functional testing, usability testing, debugging, revisions, and refinement of module	40,000
Deployment and Configuration	Hosting setup, deployment, environment configuration, and system initialization	20,000
Total Development Cost		**320,000PHP**

Operational and Maintenance Costs (Annual)

These costs represent the yearly expenses needed to maintain the system and ensure its continuous availability, reliability, and usability.

Table 2. Operational and Maintenance Costs (Annual)

Cost Component	**Annual Cost (PHP)**
Web Hosting and Domain Maintenance	30,000
System Maintenance and Bug Fixes	24,000
Security, Backup, and Monitoring	18,000
User Training and Technical Support	12,000
Total Annual Operational Cost	**84,000 PHP**

Total Cost Over 3 Years

Total Cost = Development Cost + (3 × Annual Operational Cost)

Total Cost = 320,000 + (3 × 84,000)

Total Cost = 572,000 PHP

Benefit Analysis

A. Tangible Benefits (Annual)

The proposed system is expected to generate measurable benefits through reduced manual processing time, improved transaction efficiency, lower paper-based expenses, and better use of administrative effort. These benefits are aligned with the broader view that digitization reduces administrative burden by lowering the time and effort required to complete routine processes.

Table 3. Tangible Benefits (Annual)

Benefit Category	Description	Annual Savings (PHP)
Administrative Time Savings	Less time spent on manual encoding, membership validation, record retrieval, and repetitive checking of student organization data	110,000 PHP
Improved Processing Efficiency	Faster handling of announcements, event coordination, organization requests, and member-related transactions	60,000 PHP
Reduced Paper and Printing Expenses	Lower use of printed forms, notices, attendance sheets, and manual reports through digitized processing	25,000 PHP
Reduced Repetitive Follow-up Work	Fewer duplicate submissions, manual clarifications, and repeated status follow-ups because data is centralized and easier to track	40,000 PHP

Better Utilization of Administrative Resources	Existing personnel can devote more time to coordination, student support, and decision-making instead of routine clerical tasks	35,000 PHP
Total Tangible Benefits (Annual)		**270,000 PHP**

B. Intangible Benefits

Although not directly quantifiable in monetary terms, the proposed system is expected to provide significant institutional and operational advantages. It will improve accessibility to organization-related services and information, allowing users to easily access what they need anytime. The system will also increase student engagement and participation in organizational activities by providing a more interactive and user-friendly platform. Communication among student members, officers, advisers, and administrators will become more efficient and streamlined. In addition, the system will enhance the accuracy, transparency, and accountability of organization records and transactions. It will enable faster access to information, supporting effective monitoring, reporting, and decision-making processes. Users will also experience greater convenience and satisfaction through a centralized web-based platform. Overall, the system will contribute to stronger institutional support for digitalization efforts and the modernization of services. These types of benefits are consistent with research on digital public services and GovTech, which emphasizes improved access, better service

delivery, and more efficient use of resources through digital systems.

C. Total Benefits Over 3 Years

The total benefits of the proposed system over a three-year period are derived from the annual tangible benefits that the system is expected to generate. These benefits may include cost savings, improved efficiency, reduced manual workload, and other measurable financial gains brought about by the implementation of the system. Given that the estimated annual tangible benefit is 270,000 PHP, the total benefit over three years is calculated by multiplying this value by three. As a result, the system is projected to generate a total of 810,000 PHP in tangible benefits over the three-year period. This cumulative benefit reflects the long-term value of the system and highlights its capacity to continuously deliver measurable advantages to the organization over time.

Total Benefits = 3 × Annual Tangible Benefits

Total Benefits = 3 × 270,000 = 810,000 PHP

D. Net Benefit Computation

The net benefit represents the overall financial gain that the organization will achieve after accounting for all system-related costs. It is calculated by subtracting the total system cost from the total benefits over the same period. In this case, the total system cost over three years amounts to 572,000 PHP, while the total benefits reach 810,000 PHP. By deducting the total cost from the total benefits (810,000 −

572,000), the resulting net benefit is 238,000 PHP. This positive net benefit indicates that the system not only recovers its costs but also generates additional financial value for the organization. It demonstrates that the investment is worthwhile and contributes to the overall financial sustainability and efficiency of operations.

Net Benefit = Total Benefits − Total Costs

Net Benefit = 810,000 − 572,000 = 238,000 PHP

E. Return on Investment (ROI)

Return on Investment (ROI) is a key financial metric used to evaluate the efficiency and profitability of an investment. It measures how much return is gained relative to the cost of the investment and is expressed as a percentage. The ROI is computed by dividing the net benefit by the total system cost and then multiplying the result by 100. Using the given values, the computation is as follows: (238,000 / 572,000) × 100, which results in an ROI of 41.61%. This means that for every peso invested in the system, the organization can expect a return of approximately 41.61 centavos over the three-year period. A positive ROI of this magnitude indicates that the system is financially beneficial and justifies the investment, as it provides a significant return compared to the costs incurred.

ROI = (Net Benefit / Total Cost) × 100

ROI = (238,000 / 572,000) × 100 = 41.61%

F. Cost-Benefit Summary

Table 4. Cost-Benefit Summary

Metric	Value
Total Development Cost	320,000 PHP
Total 3-Year Operational Cost	252,000 PHP
Total System Cost (3 Years)	572,000 PHP
Total Benefits (3 Years)	810,000 PHP
Net Benefit	238,000 PHP
ROI	41.61%

The overall cost-benefit analysis presents a comprehensive view of the financial implications of implementing the proposed system. The total development cost is estimated at 320,000 PHP, which covers expenses related to system design, development, testing, and initial deployment. In addition, the system incurs operational costs amounting to 252,000 PHP over three years, which may include maintenance, updates, hosting, and technical support. When combined, these costs result in a total system cost of 572,000 PHP for the entire three-year period.

On the other hand, the system is expected to generate total benefits of 810,000 PHP over the same timeframe. This leads to a net benefit of 238,000 PHP,

indicating that the system produces more value than it consumes in resources. Furthermore, with a calculated ROI of 41.61%, the analysis clearly demonstrates that the proposed system is not only cost-effective but also a profitable investment.

Overall, the cost-benefit summary highlights the financial feasibility and sustainability of the system. It supports the conclusion that implementing the proposed solution will bring long-term economic advantages, improve operational efficiency, and contribute positively to the organization's strategic goals.

Conclusion

The cost-benefit analysis shows that the proposed Integrated Management System Enhancing Student Organization Members' Accessibility and Engagement is economically feasible within a three-year evaluation period. The estimated total benefits exceed the projected development and maintenance costs, resulting in a positive net benefit and a favorable return on investment.

The tangible gains of the proposed system do not depend on reducing the number of administrative personnel. Instead, the system is expected to reduce manual effort, shorten processing time, lessen repetitive follow-up work, and improve the efficiency of existing administrative processes. This means that the value of the system lies in better productivity, improved service delivery, and more effective use of available resources. That approach is consistent with administrative-simplification and digital-service literature, which treats lower

burden and better efficiency as core benefits of digitization.

Beyond measurable financial benefits, the system also supports stronger transparency, accessibility, communication, and student engagement in organization-related activities. These benefits further justify the implementation of the proposed system in the National Aviation Academy of the Philippines – Villamor as a practical and institutionally beneficial solution.

System Requirements Specification

This section presents the System Requirements Specification of the proposed Integrated Management System Enhancing Student Organization Members Accessibility and Engagement in the National Aviation Academy of the Philippines – Villamor. It provides a structured presentation of the system requirements necessary for the development and implementation of the proposed web-based platform. Specifically, this section identifies the relevant stakeholders and their corresponding user roles, defines the functional requirements that describe the essential services and operations the system must perform, and outlines the non-functional requirements that establish the standards of quality needed for effective system operation. Through these specified requirements, the study ensures that the proposed system remains aligned with the institutional objectives of improving accessibility, strengthening administrative efficiency, supporting student organization management, and providing a secure, reliable,

and user-centered digital environment.

Stakeholder Identification and User Roles

The proposed Integrated Management System Enhancing Student Organization Members Accessibility and Engagement in the National Aviation Academy of the Philippines – Villamor involves several key stakeholder groups whose participation is necessary for the effective implementation and operation of the system. Identifying these stakeholders helps ensure that the system's functions are aligned with institutional workflows, governance requirements, and role-based access controls. Since the proposed platform is intended to centralize student organization management, document handling, monitoring, and service accessibility, clearly defined user roles are essential to maintain security, accountability, and operational efficiency.

Students constitute one of the primary stakeholder groups of the system. As intended end-users, they access organization-related information, receive notifications, monitor activities, and engage with services made available through the web-based platform. The system is designed to improve their accessibility to announcements, organizational records, and participation-related information through a centralized and user-friendly environment. Their role in the system is mainly centered on viewing relevant information, receiving updates, and interacting with services that support student organization engagement within the

institution.

Student organization officers serve as another major stakeholder group because they are directly involved in the day-to-day management of organizational activities and records. Through the system, they are expected to manage membership-related information, submit required organizational documents, monitor attendance and activities, and coordinate with the Office of Student Affairs through digital workflows. Their role is more operational in nature, as they act as the main organizational users responsible for maintaining updated records and complying with institutional submission and reporting requirements.

Office of Student Affairs (OSA) administrators function as the primary oversight and approval stakeholders within the proposed system. As the office responsible for supervising student organizations, OSA administrators are expected to review submitted documents, monitor compliance, oversee organizational transactions, and use dashboard and reporting tools for decision-making and administrative control. Their access to the system supports more efficient monitoring, better coordination, and improved institutional oversight compared with the current manual and fragmented process. This role is especially important because the study emphasizes governance efficiency, regulatory compliance, and administrative effectiveness in student organization management.

System administrators also play a critical role in ensuring the stability, security, and continuous operation of the platform. Their responsibilities include managing user accounts, maintaining access permissions, safeguarding stored organizational and student data, and ensuring that the system remains functional, secure, and aligned with institutional requirements. Since the proposed system includes secure authentication, role-based access control, and a centralized database, administrative management of technical operations is necessary to preserve data integrity and maintain the reliability of digital services.

Taken together, the identification of these stakeholder groups strengthens the overall design and usability of the proposed system. Role-based access ensures that students, student organization officers, OSA administrators, and system administrators interact only with functions relevant to their responsibilities. This improves security, supports accountability, and promotes more efficient execution of student organization processes. In the context of the study, clearly defined user roles also contribute to the successful implementation of the RAD-based development approach because continuous stakeholder involvement helps ensure that the final system remains responsive to actual institutional needs.

Functional Requirements

The proposed Integrated Management System Enhancing Student Organization Members Accessibility and Engagement in the National Aviation Academy of the Philippines – Villamor shall provide the core functions needed to

support centralized, efficient, and secure student organization management. These functional requirements define the major system features that will improve accessibility of services, document handling, communication, monitoring, and administrative control within the institution.

To ensure secure access, the system shall provide a login and authentication feature for all authorized users. Students, student organization officers, and administrators shall log in using valid credentials before accessing the system. The platform shall also implement role-based access control so that each user can only access the features and data relevant to their assigned responsibilities.

In support of organization processes, the system shall include a document submission and approval function that allows student organization officers to submit required files digitally and enables administrators to review, verify, and monitor these submissions. The system shall also provide an organization and membership management feature for maintaining organization records, profiles, and related information in a centralized database.

To improve coordination and service delivery, the system shall include a notification feature that sends updates, reminders, and announcements to relevant users regarding activities, submissions, and other transactions. In addition, the platform shall provide dashboard and reporting features that allow administrators to monitor records, track organizational activities, and support decision-making

through organized and accessible information.

The system shall also maintain a centralized database for storing user data, organization records, submitted documents, and transaction histories. This shall support more accurate record retrieval, improved monitoring, and better records consolidation compared to the current manual process.

These functional requirements support the development of a secure, centralized, and user-oriented web-based platform for student organization management at NAAP–Villamor. By integrating authentication, role-based access, digital workflows, notifications, reporting, and centralized record management, the proposed system is expected to improve operational efficiency, transparency, accessibility, and administrative effectiveness.

Non Functional Requirements

The proposed Integrated Management System Enhancing Student Organization Members Accessibility and Engagement in the National Aviation Academy of the Philippines – Villamor shall meet essential quality standards to ensure that the system operates effectively and supports the institutional needs of students, student organization officers, and administrators. These non-functional requirements focus on the overall quality of the system in terms of security, usability, reliability, performance efficiency, maintainability, compatibility, portability, and accessibility.

In terms of security, the system shall protect user accounts, organizational records, and submitted documents through secure authentication and role-based access control. Only authorized users shall be allowed to access specific features and data relevant to their responsibilities. This is necessary to maintain confidentiality, accountability, and protection of institutional information.

With respect to usability, the system shall provide a clear, organized, and user-friendly interface that enables users to perform tasks easily and efficiently. Navigation, forms, and system processes shall be designed in a way that supports convenient access to records, submissions, notifications, and reports. This requirement is important to ensure that the platform remains practical and understandable for all intended users.

For reliability and performance efficiency, the system shall maintain stable operation and acceptable response time during regular use. It shall be capable of processing submissions, approvals, and notifications without unnecessary delays. The system shall also preserve data accuracy and support dependable record storage to reduce interruptions and maintain smooth institutional operations.

In relation to maintainability, the platform shall be structured in a way that allows easier correction of errors, updating of features, and future enhancement of services. The use of the Rapid Application Development (RAD) model supports this requirement by encouraging iterative refinement, testing, and continuous improvement throughout development.

As for compatibility, portability, and accessibility, the system shall function properly across standard web browsers and commonly used devices within the institution. It shall remain usable in the institution's web-based environment and support convenient access to services and records for authorized users. These qualities are necessary to ensure that the proposed system can be implemented effectively and sustained over time.

These non-functional requirements support the development of a secure, reliable, efficient, and user-centered platform for student organization management at NAAP–Villamor. By meeting these quality standards, the proposed system can better support accessibility, streamline administrative workflows, and improve the overall delivery of student organization services.

Proposed System Architecture and Design

The proposed Integrated Management System Enhancing Student Organization Members Accessibility and Engagement in the National Aviation Academy of the Philippines – Villamor is designed as a web-based and centralized platform intended to improve the management of student organization records, services, and administrative processes. Its architecture and design were developed to provide a structured, secure, and organized environment that supports accessibility, operational efficiency, and effective coordination among students, organization officers, and OSA administrators/staff. Through an integrated system framework, the proposed platform establishes the foundation for

the presentation of its overall architecture and the major system components and modules that enable its operation.

System Architecture Overview

The proposed Integrated Management System Enhancing Student Organization Members Accessibility and Engagement in the National Aviation Academy of the Philippines – Villamor adopts a web-based layered system architecture designed to support centralized data management, secure system access, and efficient delivery of student organization services. The architecture enables multiple stakeholders, including students, student organization officers, and OSA administrators/staff, to interact with the system through a web browser or mobile-responsive user interface, while maintaining coordinated application processing, dashboard-based monitoring and reporting, and centralized database storage.

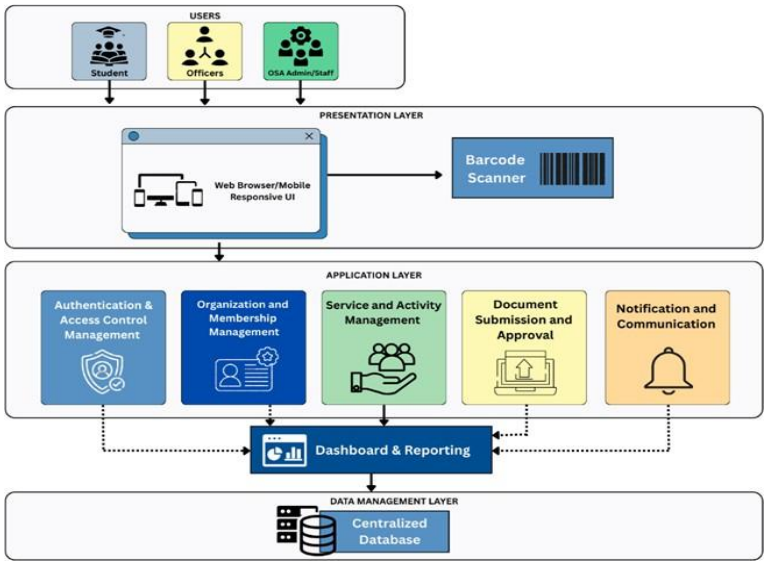

Figure 2. System Architecture of the Integrated Management System Enhancing Student Organization Members and Accessibility and Engagement in National Aviation Academy of the Philippines - Villamor

The proposed Integrated Management System Enhancing Student Organization Members Accessibility and Engagement in the National Aviation Academy of the Philippines – Villamor is organized into three major architectural layers, namely the presentation layer, application layer, and data management layer. This layered structure is intended to support secure user interaction, efficient processing of system functions, and centralized storage of organizational data. Through this architecture, the system provides an organized framework for managing student organization services, document transactions, notifications, reporting, and administrative processes within a unified web-based environment.

At the upper level, the presentation layer serves as the user-facing component of the system. It provides access through a web browser or mobile-

responsive interface, allowing students, organization officers, and OSA administrators/staff to interact with the platform using standard internet-enabled devices. This layer is designed to support convenient navigation, accessibility, and responsiveness across different screen sizes, making it easier for users to submit documents, manage records, access services, and monitor updates. The inclusion of a barcode scanner in the interface layer also supports faster identification and processing of selected transactions within the system.

At the center of the architecture, the application layer functions as the main processing environment of the system. This layer contains the core modules responsible for authentication and access control management, organization and membership management, service and activity management, document submission and approval, notification and communication, and dashboard and reporting. These modules work together to execute the primary functions of the platform, including secure login, role-based access, handling of organization records, digital submission workflows, delivery of notifications, and generation of reports. The application layer also ensures that each user interacts only with the features relevant to their assigned role, thereby promoting security, accountability, and proper control of system operations.

At the lower level, the data management layer is responsible for maintaining the centralized database of the system. This layer stores user accounts, organization profiles, membership records, submitted documents, communication

logs, activity data, and other related transactions necessary for system operation. Centralized storage supports consistency of records, more efficient retrieval of information, and better coordination among system modules. It also strengthens reporting and monitoring functions by ensuring that all processed data are maintained within a single and organized repository.

The layered design supports efficient deployment and continuous operation of the proposed platform. Because the system is web-based, multiple users can access its services at the same time while administrators manage records, approvals, and monitoring tasks from a centralized environment. This setup also makes the system easier to maintain, update, and expand as institutional needs change. The arrangement of modules and database support further contributes to smoother integration of student organization processes and improved digital service delivery.

The proposed system architecture provides a secure, scalable, and organized framework for student organization management at NAAP–Villamor. By combining a responsive user interface, integrated application modules, dashboard-based monitoring, and centralized data storage, the architecture supports improved accessibility, administrative efficiency, and more effective handling of student organization records and services.

System Components and Modules

The proposed Integrated Management System Enhancing Student Organization Members Accessibility and Engagement in the National Aviation Academy of the Philippines – Villamor is composed of several interconnected modules designed to support centralized student organization management, digital document processing, administrative monitoring, and improved access to organization-related services and information. The modular structure of the system promotes organized system development, easier maintenance, and efficient coordination of institutional processes. Through this design, the system addresses the limitations of manual submissions, fragmented communication, and inconsistent record management by integrating key functions into a unified web-based platform.

The Authentication and Access Control Module serves as the security gateway of the system. This component manages user login, verifies authorized access, and applies role-based permissions for students, student organization officers, and Office of Student Affairs administrators. By limiting access according to user responsibilities, the module helps protect records and transactions while supporting accountability, confidentiality, and secure system use.

The Organization and Membership Management Module functions as one of the core operational parts of the platform. This module maintains organization profiles, membership-related information, participation records, and other essential

organizational data within a centralized database. It enables student organization officers and authorized administrators to manage records more efficiently while reducing redundancy and improving accessibility of information.

The Document Submission and Approval Module manages the digital processing of organizational requirements. Through this module, student organization officers can upload reports and other required documents, while Office of Student Affairs administrators can review, verify, approve, and monitor submissions through a structured workflow. This component supports compliance with streamlined and electronic transaction practices by reducing reliance on physical paperwork and improving transparency in document handling.

The Notification and Communication Module is intended to improve coordination among users of the system. This module delivers updates, reminders, announcements, and transaction status notifications related to submissions, approvals, organization activities, and student requests. Its inclusion helps minimize missed communications, reduce repeated follow-ups, and strengthen responsiveness among students, officers, and administrators. Related literature also supports the value of centralized notification channels in improving accountability and institutional coordination.

The Dashboard and Reporting Module provides a consolidated view of organizational information, submissions, and monitored activities. Through this component, administrators can track compliance, review records, observe activity

trends, and support decision-making using organized and accessible system data. This module strengthens institutional oversight by transforming stored records into useful reports and monitoring tools that enhance governance efficiency and administrative effectiveness.

The Service and Activity Management Module supports student engagement by allowing users to access organization-related services, monitor participation in activities, and track requests or transactions associated with student organizations. This module reflects the study's goal of improving accessibility and engagement by giving students a more convenient way to interact with organization services, events, and related processes through a single platform.

The Administrative Management Module oversees the overall operation of the platform. Its functions include user account administration, monitoring of system activity, management of institutional records, and maintenance of system settings necessary for continuous operation. This module helps ensure that the system remains organized, secure, and aligned with institutional workflows and operational requirements.

Supporting these modules is the Centralized Database Component, which stores user information, organization records, submitted documents, transaction histories, and activity logs. This component serves as the primary repository of system data and is essential in maintaining data accuracy, record consolidation,

and efficient retrieval of organizational information. The use of a centralized database also supports monitoring, reporting, and secure records management throughout the platform.

Taken together, these system components enable the proposed platform to operate as a secure, centralized, and user-oriented management system for NAAP–Villamor. The modular architecture supports maintainability, scalability, and efficient integration of student organization processes while remaining consistent with the study's objectives and the RAD-based SDLC adopted for system development.

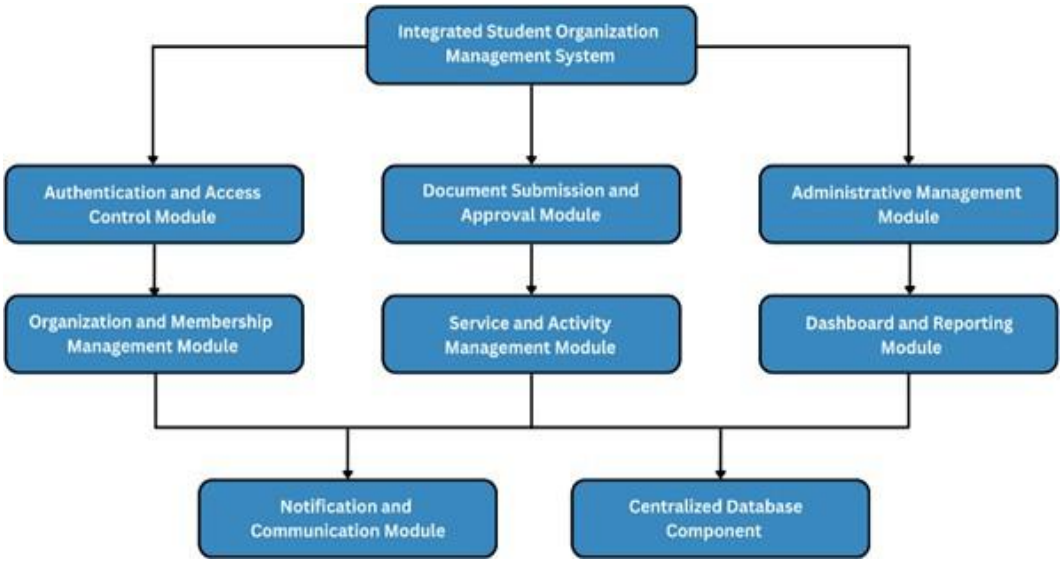

Figure 3. Proposed Modular Architecture of the Integrated Management System Enhancing Student Organization Members and Accessibility and Engagement in National Aviation Academy of the Philippines - Villamor

Figure 3 presents the modular architecture of the proposed Integrated Student Organization Management System. The Integrated Student Organization Management System serves as the core platform that coordinates the major system functions and supports the centralized management of student organization processes. Connected to the core are the Authentication and Access Control Module, which ensures secure user login and role-based access permissions, the Document Submission and Approval Module, which manages the digital submission, review, and approval of organizational requirements, and the Administrative Management Module, which oversees system administration, user management, and operational control. Supporting these primary modules are the Organization and Membership Management Module, which handles organization profiles, membership records, and related organizational data, the Service and Activity Management Module, which manages student organization services, activities, and participation-related processes, and the Dashboard and Reporting Module, which provides consolidated records, monitoring tools, and reports for administrative decision-making. At the lower level of the architecture are the Notification and Communication Module, which delivers announcements, updates, and transaction-related notifications, and the Centralized Database Component, which serves as the main repository of system data, records, and transaction histories. The modular and layered arrangement of these components supports

scalability, maintainability, clarity of process flow, and efficient integration of the proposed system.

Process Modeling and System Workflow

This section presents the process modeling and system workflow of the proposed Integrated Management System Enhancing Student Organization Members Accessibility and Engagement in the National Aviation Academy of the Philippines – Villamor. It provides a structured representation of how the system operates by illustrating its overall system boundary, the flow of data between external stakeholders and internal processes, the interaction of major processing components and data stores, and the sequence of activities performed during system use. Through the context diagram, Data Flow Diagram (Level 0), Data Flow Diagram (Level 1), and procedural workflow, this section explains how the proposed platform manages user access, organization-related transactions, document handling, event and attendance processes, rental services, and administrative monitoring within a centralized and organized web-based environment.

Context Diagram

The context diagram presents the overall system boundary and the interactions between the proposed Integrated Management System Enhancing Student Organization Members Accessibility and Engagement in the National

Aviation Academy of the Philippines – Villamor and its external stakeholders. In the diagram, the system is treated as a single centralized process that exchanges information with the primary users involved in student organization management at NAAP–Villamor, particularly students, student organization officers, Office of Student Affairs (OSA) administrators, and system administrators. This high-level view emphasizes the system's role as a unified web-based platform for improving accessibility, coordination, document handling, monitoring, and administrative efficiency.

Students interact with the system by accessing organization-related information, receiving notifications, monitoring activities, and using organization-related services made available through the platform. Student organization officers interact with the system by managing organization and membership records, submitting required documents, monitoring activities, and coordinating transactions through digital workflows. These interactions reflect the study's goal of improving student access to services, records, requests, and updates through a centralized platform.

OSA administrators engage with the system by reviewing submitted documents, monitoring compliance, overseeing organization-related transactions, and using dashboard and reporting tools for decision-making and administrative control. System administrators, on the other hand, interact with the platform by managing user accounts, maintaining access permissions, monitoring system

activity, and ensuring the security and continuous operation of the platform. These roles support the study's emphasis on stronger oversight, secure access, and centralized management of student organization processes.

The context diagram does not present the internal modules or detailed processing logic of the system. Instead, it focuses only on the external entities and the major data exchanges that occur between them and the proposed platform, such as submissions, approvals, notifications, records, monitoring information, and administrative controls. This representation is important because it helps define the scope of the system, identify stakeholder participation, and clarify the flow of information needed for the design and development of the proposed web-based student organization management system.

Figure 4. Context Diagram of the Integrated Management System Enhancing Student Organization Members and Accessibility and Engagement in National Aviation Academy of the Philippines – Villamor

Figure 4 illustrates the context diagram of the proposed Integrated Student Organization Management System. The diagram presents the system as a centralized process that interacts with external stakeholders, including students, student organization officers, and OSA administrators. It highlights the exchange of organization-related information, document submissions, activity and service records, notifications, monitoring data, and administrative control information between the users and the system, thereby defining the system boundary and the scope of stakeholder interaction.

Data Flow Diagram (Level 0)

The Level 0 Data Flow Diagram presents the major data exchanges of the proposed Integrated Student Organization Management System and shows how the system interacts with its primary external users. In the diagram, the system serves as the central platform that receives, processes, and returns information related to registration, login, document submission, approvals, announcements, attendance, and rental transactions. This representation provides a high-level view of how information moves between the system and the stakeholder groups involved in student organization activities and related services.

From the perspective of students, the system accepts data such as registration details, login credentials, document submissions, and rental-related requests. In return, the system provides outputs including approvals, announcements, and rental status updates. This flow indicates that students use the platform not only for account access but also for receiving official updates and monitoring the status of their transactions within the organization management process.

In relation to organization officers, the system receives inputs associated with registration, login, event setup, attendance scans, announcements, document submissions, and rental processing. After processing these transactions, the system returns information such as attendance confirmations, repository status, and inventory updates. This shows that organization officers have a more

operational role in the system because they handle a wider range of organization-related activities and records.

For OSA administrators, the diagram shows that the system supports administrative and oversight functions. The system receives inputs involving registration, login, approvals, policies, and campus announcements, while it provides outputs such as registration decisions, oversight dashboards, and document submission information. This data flow reflects the role of OSA administrators in supervising organization processes, reviewing transactions, and monitoring institutional compliance through the platform.

As shown in the diagram, event attendees or guests interact with the system mainly through barcode scans or attendance data, while the system returns attendance confirmation when applicable. This exchange supports event monitoring and participant validation. In the same way, renters interact with the platform by submitting rental requests and returns, while the system provides item hand-off instructions and fee-related information. These flows demonstrate that the system also functions as a support platform for event participation and rental-related transactions beyond standard organization records management.

The Level 0 Data Flow Diagram illustrates the system as a centralized digital platform that connects multiple stakeholder groups through structured data exchange. It defines the broad scope of system interaction without presenting lower-level internal processing details, thereby helping clarify how the proposed

Integrated Student Organization Management System manages organization-related information, approvals, attendance, announcements, and rental transactions in a unified environment.

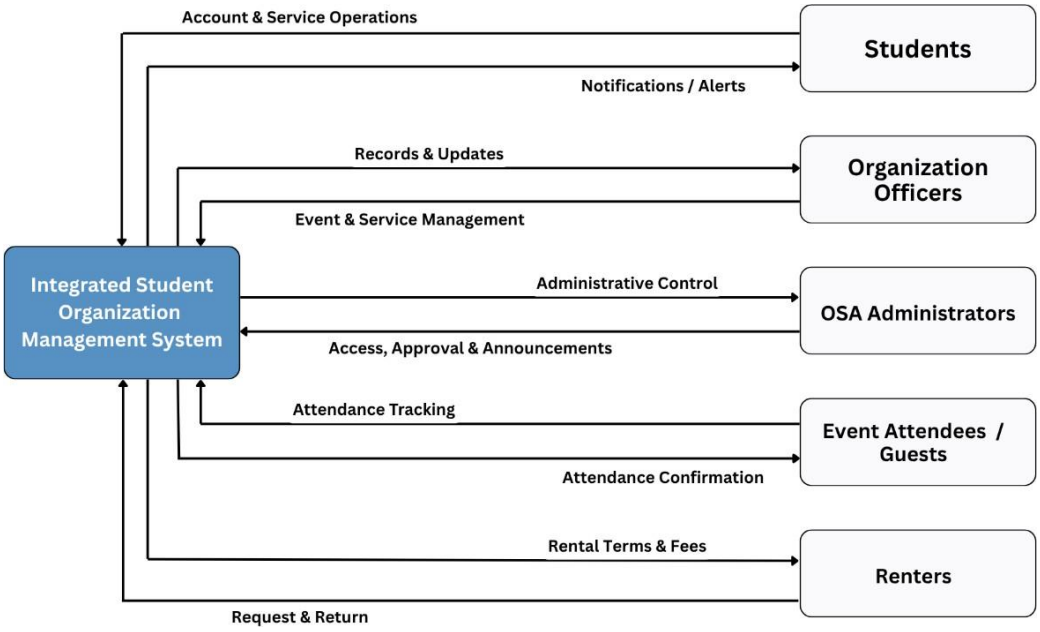

Figure 5. Data Flow Diagram (Level 0) of the Integrated Management System Enhancing Student Organization Members and Accessibility and Engagement in National Aviation Academy of the Philippines - Villamor

Figure 5 illustrates the Level 0 Data Flow Diagram of the proposed Integrated Student Organization Management System. The diagram presents the system as a centralized process that interacts with external stakeholders, including students, organization officers, OSA administrators, event attendees or guests, and renters. It highlights the exchange of registration and login data, document submissions, approvals, announcements, attendance information, rental transactions, and administrative control data between the users and the system,

thereby defining the scope of interaction and the major flow of information within the proposed platform.

Data Flow Diagram Level 1

The Level 1 Data Flow Diagram presents a more detailed view of the internal processes of the proposed Integrated Student Organization Management System by breaking down the major functions shown in the Level 0 DFD into more specific processing components. Through this diagram, the flow of data between the system processes, data stores, and external entities becomes clearer. It shows how the platform handles user account transactions, event and attendance activities, document-related processes, and rental operations while maintaining coordination with students, organization officers, OSA administrators, event attendees or guests, and renters.

At the first level of processing, the Manage Users and Accounts (1.0) process handles registration and login transactions submitted through the system. This process is responsible for receiving account-related inputs, validating user access, and maintaining account information through the User Accounts DB (D1). It supports the system's need for controlled access by ensuring that users are properly registered and authenticated before they can proceed with other transactions. Because students and other stakeholders begin their system interaction through registration and login, this process serves as one of the primary

entry points of the platform.

The second process, Event Setup and Attendance Processing (2.0), focuses on event-related activities managed within the system. It receives data associated with event setup and attendance operations, particularly from organization officers and attendance-related transactions. The process stores and retrieves relevant records using the Events Repository (D2). Through this component, the system is able to generate outputs such as attendance confirmations, repository status, and inventory-related updates, which support the operational responsibilities of organization officers and the monitoring functions required by the institution.

In a similar manner, the Document Repository Processing (3.0) component manages document submission transactions and related record handling. This process accepts document inputs, supports the review and routing of submitted files, and stores the processed records in the Document File Store (D3). It also contributes to the production of outputs needed by OSA administrators, such as document submission information, registration decisions, and oversight-related data. This part of the diagram highlights the role of the system in organizing document workflows and maintaining a centralized repository of institutional files associated with student organization activities.

Another important process shown in the diagram is Manage Rentals and Inventory (4.0), which deals with rental transactions and inventory-related records.

The system receives rental requests, returns, and processing inputs, then manages these transactions using the Inventory and Rentals DB (D4). As a result, the system can provide outputs such as item hand-off instructions and fee-related information to renters. This process indicates that the proposed platform is not limited to record management and document handling alone, but also supports operational services connected to organizational resources and rental activities.

The Level 1 diagram also illustrates how external entities interact with the internal processes of the system. Students primarily exchange data related to registration, login, document submissions, rentals, approvals, announcements, and rental status. Organization officers are associated with event setup, attendance scans, announcements, document submissions, and rental processing, while also receiving attendance confirmations, repository status, and inventory updates. OSA administrators interact with the system through approval-related and oversight-related transactions, while event attendees or guests participate through barcode scans or attendance data and receive attendance confirmation when applicable. Renters communicate with the system through rental requests and returns, then receive instructions and fee-related outputs after processing.

The Level 1 Data Flow Diagram provides a more specific representation of how the proposed Integrated Student Organization Management System processes information internally. It demonstrates how the platform distributes

transactions across key functional processes and corresponding data stores while maintaining consistent interaction with external users. By presenting the detailed movement of account data, event information, documents, attendance records, and rental transactions, the diagram strengthens the understanding of how the proposed system operates as an integrated and centralized digital platform for student organization management.

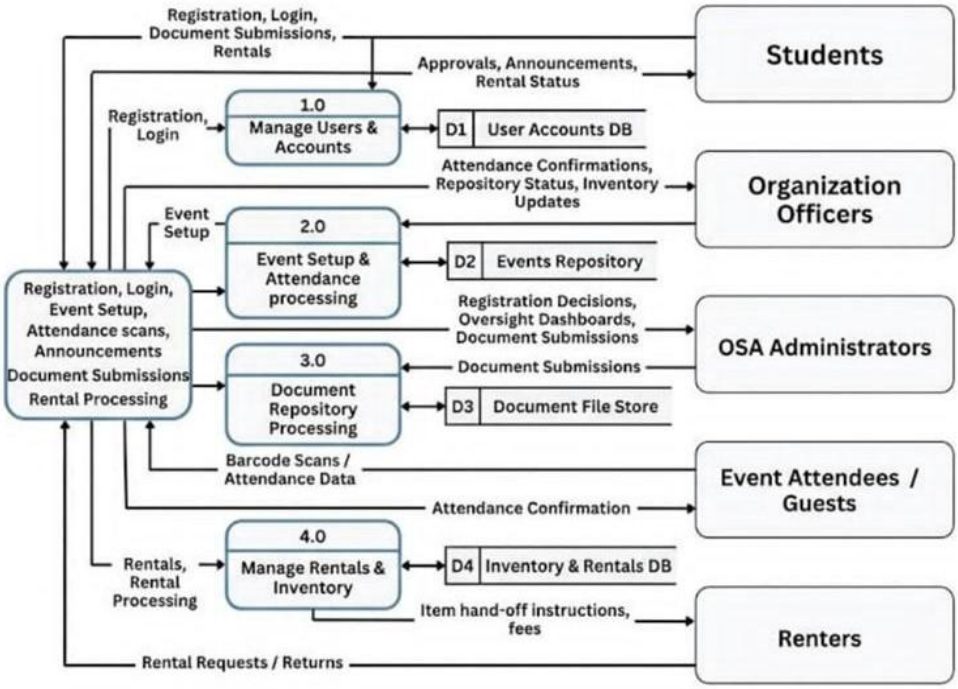

Figure 6. Data Flow Diagram (Level 1) of the Integrated Management System Enhancing Student Organization Members and Accessibility and Engagement in National Aviation Academy of the Philippines – Villamor

Figure 6 presents the Level 1 Data Flow Diagram of the proposed Integrated Student Organization Management System. The diagram illustrates the detailed internal processes of user and account management, event setup and attendance

processing, document repository processing, and rental and inventory management. It also shows the exchange of data among external stakeholders, system sub-processes, and the corresponding data stores, including the User Accounts DB, Events Repository, Document File Store, and Inventory and Rentals DB, thereby providing a more detailed representation of the internal operations of the proposed system.

Procedural Workflow

The procedural workflow of the proposed Integrated Student Organization Management System describes the sequence of activities performed by the users and the corresponding system responses during the management of student organization transactions. It explains how information is entered, processed, stored, and returned through the platform to support registration, login, event setup, attendance processing, document submission, announcements, approvals, rental transactions, and administrative monitoring. Through this workflow, the system provides a centralized and organized process for handling the major operations related to student organization management.

At the beginning of the procedure, the user accesses the system through the web-based platform and submits the required registration or login information. The system then processes these inputs through the Manage Users and Accounts function, where the data are validated and checked against the User Accounts

Database. If the submitted information is valid, the user is granted access to the system according to the assigned role. This step ensures that students, organization officers, OSA administrators, and other authorized users are directed only to the functions relevant to their responsibilities.

After successful access, the user proceeds to the appropriate transaction based on the assigned role and current activity. For students, the workflow may involve document submission, viewing announcements, checking approval status, or initiating rental-related transactions. For organization officers, the procedure may include event setup, attendance scanning, posting announcements, submitting organizational documents, and processing rentals. For OSA administrators, the workflow involves reviewing registrations, monitoring submitted files, issuing approvals, managing policies, and viewing oversight dashboards. These activities serve as the main operational inputs received by the system.

Once a transaction is initiated, the system routes the data to the corresponding internal process. Registration and login requests are handled by Manage Users and Accounts. Event-related and attendance-related activities are processed through Event Setup and Attendance Processing, which stores and retrieves relevant information from the Events Repository. Document-related transactions are directed to Document Repository Processing, where submitted files are recorded and maintained in the Document File Store. Rental requests, returns, and related inventory activities are managed through Manage Rentals and

Inventory, which uses the Inventory and Rentals Database for storage and retrieval. In this stage, the system validates, updates, and organizes the information according to the nature of the transaction.

As processing continues, the system generates the necessary outputs for the user concerned. Students receive outputs such as approvals, announcements, and rental status updates. Organization officers obtain attendance confirmations, repository status, and inventory updates. OSA administrators receive registration decisions, oversight dashboards, and document-related information needed for administrative control. Event attendees or guests may receive attendance confirmation when barcode scans or attendance data are successfully processed. Renters, on the other hand, receive item hand-off instructions and fee-related information after their rental transactions are completed. These outputs ensure that each stakeholder receives timely and relevant information from the system.

At the same time, the processed data are stored in the appropriate data repositories to support record consistency, retrieval, and future monitoring. Account-related data are kept in the User Accounts DB, event and attendance information are stored in the Events Repository, submitted files are maintained in the Document File Store, and rental and inventory records are managed in the Inventory and Rentals DB. This organized data storage allows the system to maintain accurate records, support administrative monitoring, and provide updated information whenever needed.

The procedural workflow demonstrates how the proposed system coordinates user access, transaction handling, internal processing, data storage, and output generation within a single platform. By integrating account management, event and attendance processing, document repository handling, and rental and inventory control, the system supports a more efficient, accessible, and structured approach to student organization management at NAAP–Villamor.

Data Modeling and Database Design

This section presents the data modeling and database design of the Integrated Student Organization Management System. It explains how institutional data is structured, organized, and interconnected to support governance, event monitoring, documentation control, and rental services within a centralized web-based platform. The section begins with the conceptual data model, which defines the major entities and their relationships at a high level, followed by the Entity Relationship Diagram (ERD) that details the logical database structure, including keys, attributes, and relationship cardinalities. Together, these models provide a systematic foundation for implementing a reliable, scalable, and integrity-driven database architecture.

Conceptual Data Model

This conceptual data model presents the high-level data structure of the Integrated Student Organization Management System by identifying the major

entities involved in user management, academic structure, organization governance, event monitoring, documentation control, and rental services. The model defines how institutional data is organized and how different components interact within a centralized web-based platform, without yet specifying technical database implementation details.

At the core of the model is the Users entity, which represents all system stakeholders including students and authorized personnel. Users contain personal, academic, and account-related attributes that support identification, authentication, and activity tracking. The Student Number entity is linked to Users to ensure that each student has a unique institutional identifier. This strengthens traceability and ensures that all transactions, registrations, and records are properly associated with verified individuals.

The academic structure is represented through Institutes and Academic Program. Institutes offer academic programs, and students are enrolled in specific programs. This relationship organizes users according to their academic affiliation and supports administrative monitoring across institutional units. By structuring data this way, the system enables filtering and reporting based on institute or program classification.

Student organization governance is centered on the Organization entity. Each organization contains identifying and descriptive information and serves as the primary unit for events, announcements, rentals, and documentation. The Org

Role entity defines roles within organizations, specifying leadership structures and access privileges. These roles are assigned through Organization Member, which links Users to organizations and defines their responsibilities. The Program Org Mapping entity connects organizations to specific academic programs, formally documenting institutional alignment and organizational scope.

Event management is handled through the Event entity, which stores details of activities hosted by organizations. Students request participation through Pending Registration, and approved participation is recorded in the Attendance Record. Attendance records maintain verified logs of event participation and serve as a basis for engagement monitoring and reporting. This structure replaces manual attendance systems with a centralized digital tracking mechanism.

Communication within the system is facilitated by the Announcement entity. Organizations define, author, and issue announcements that are linked to members and visible to users. This ensures structured dissemination of official information and maintains documented communication history within the platform.

The documentation workflow is represented by Document Submission, Document Annotation, and Approved Document Report. Organizations submit documents that undergo review and annotation before approval. Document annotations allow comments, revisions, and feedback to be recorded systematically. Once approved, documents are stored and summarized into approved document reports, which include relevant organizational and academic

references. This structured approval process enhances compliance monitoring, transparency, and accountability in organization-related documentation.

A major operational component of the system is the rental service module. The Rental entity records rental transactions initiated by organizations or authorized users. Rentals are associated with Inventory Category, which groups items based on classification. Each category contains multiple Inventory Item records representing individual rentable resources. Rental transactions include items through the Rental Item entity, which specifies quantity, rates, and item references. This design enables accurate tracking of borrowed resources, monitoring of availability, and maintenance of inventory control.

The relationships among these entities establish a centralized and integrated data environment. Users are linked to academic structures, organizations, events, rentals, and documentation processes. Organizations manage events, issue announcements, submit documents, and oversee rental operations. Documentation passes through structured review and approval stages, while rental transactions are directly connected to inventory records. Attendance, membership, and program mappings provide additional layers of accountability and traceability.

Overall, the conceptual data model supports governance efficiency, regulatory compliance, centralized information access, and administrative effectiveness. By organizing system components into clearly defined entities and

structured relationships, the model provides a coherent foundation for the logical and physical design stages of the Integrated Student Organization Management System.

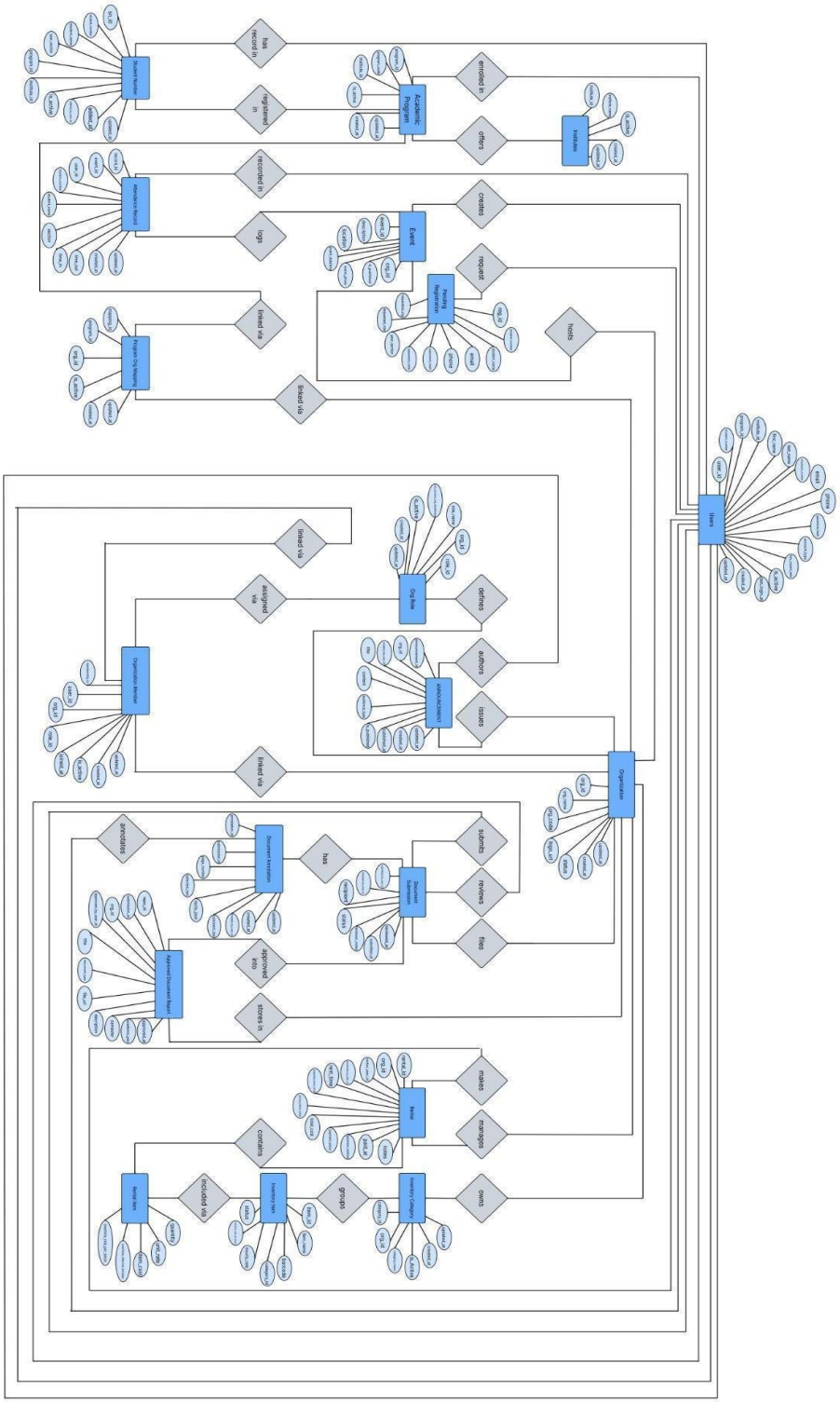

Figure 7. Conceptual Data Model

Entity Relationship Diagram

The Entity Relationship Diagram (ERD) illustrates the logical structure of the Database for the Integrated Student Organization Management System. It defines the entities, their attributes, primary keys, foreign keys, and relationship cardinalities to ensure data integrity, consistency, and controlled processing of organization governance, event management, documentation workflows, and rental services. The ERD establishes how data is interconnected within a centralized institutional platform.

The Users entity serves as the core entity of the system. It stores account and identification details such as user_id, name, email, and account_type. Users participate in multiple system processes, including creating events and announcements, submitting and reviewing documents, processing rentals, and recording attendance. The relationship between Users and other entities is primarily one-to-many, as a single user can create multiple events, announcements, rentals, or document records.

The academic structure is defined through Institutes, Academic Programs, and Student Numbers. An institute has a one-to-many relationship with academic programs, meaning one institute can offer multiple programs. Student Numbers are linked to both institutes and users, ensuring that each student record is uniquely identifiable and institutionally classified. This structure enforces referential integrity by ensuring that student records are associated with valid

academic units.

The Organizations entity represents recognized student organizations. Each organization can have multiple Organization Members, establishing a one-to-many relationship between organizations and memberships. Organization Members link users to organizations and specify roles and join dates. Organizational roles are further structured through the Org Roles entity, which defines role names within each organization. This separation ensures role standardization and prevents duplication of role definitions.

Program alignment is maintained through Program Org Mappings, which connect organizations to specific academic programs. This relationship supports institutional oversight by formally associating organizations with academic classifications.

Event management is structured around the Events entity. Each event is created by a user and belongs to one organization, forming a many-to-one relationship from events to organizations and users. Participation is tracked through Attendance Records, which reference both event_id and user_id as foreign keys. This creates a many-to-one relationship between attendance records and events, as well as between attendance records and users. The design ensures that attendance entries cannot exist without valid event and user references.

Student participation requests are handled through Pending Registrations, which store registration data linked to users and organizations. This entity supports controlled approval workflows before final participation is recorded.

The rental module is defined by Rentals, Rental Items, Inventory Categories, and Inventory Items. A rental record is associated with one organization and may involve both a renter and a processor (users), forming multiple foreign key relationships to the Users entity. Each rental can contain multiple rental items, establishing a one-to-many relationship between rentals and rental items. Rental Items reference Inventory Items, which are grouped under Inventory Categories. Inventory Categories belong to organizations, forming a one-to-many relationship between organizations and categories, and between categories and inventory items. This layered structure ensures accurate tracking of item ownership, classification, and usage while preventing orphan inventory records.

Documentation control is managed through Document Submissions, Document Annotations, and Documents Approved. A document submission is linked to the submitting and reviewing users and to a specific organization. One document submission can have multiple document annotations, forming a one-to-many relationship that supports structured review and feedback processes. Once approved, records are stored in Documents Approved, which references the original submission and the approving user. This ensures traceability from

submission to final approval and enforces accountability through foreign key constraints.

Announcements are represented by the Announcements entity. Each announcement is created by a user and associated with an organization, establishing many-to-one relationships that ensure announcements cannot exist without valid organizational and user references.

Throughout the ERD, primary keys uniquely identify each entity, while foreign keys enforce referential integrity across related entities. Cardinalities such as one-to-many and many-to-many (resolved through associative entities like Organization Member and Rental Items) ensure structured data relationships and prevent duplication or inconsistency.

By defining structured entity relationships, primary and foreign key constraints, and appropriate cardinalities, the ERD supports reliable data storage, efficient reporting, controlled documentation workflows, accurate inventory tracking, and centralized governance monitoring. The diagram provides the logical foundation for implementing a consistent and scalable database design for the Integrated Student Organization Management System.

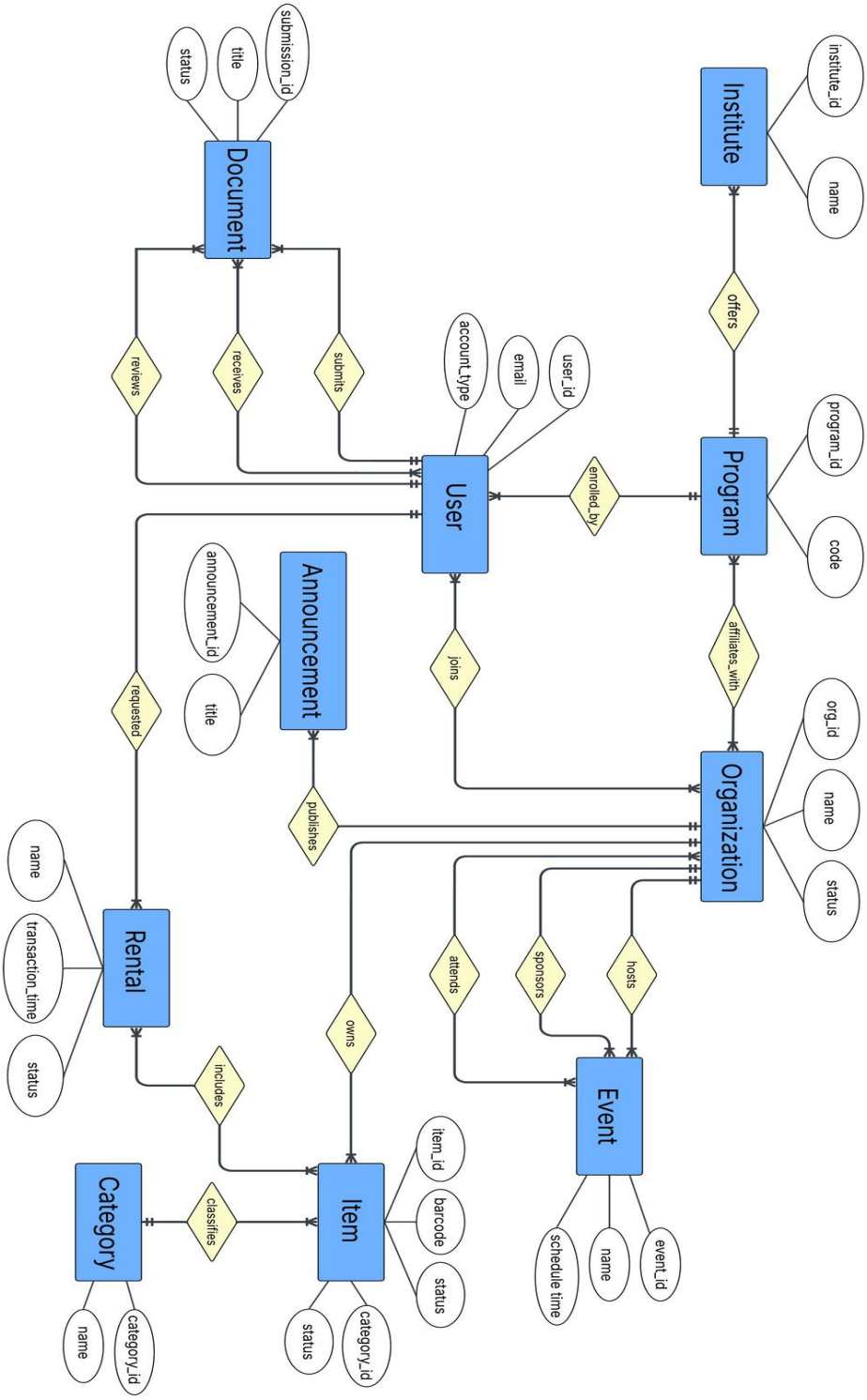

Figure 8. Entity Relationship Diagram

Logical Database Design

The logical database design defines the structured organization of data within the Integrated Student Organization Management and Rental System. The logical model translates the conceptual framework into relational database tables by specifying attributes, primary keys, foreign keys, and relational constraints. This structure ensures data consistency, integrity, and efficient processing of student organization governance, document workflows, event management, attendance monitoring, and rental services within the institution.

The database is composed of multiple interrelated tables grouped into core functional modules: user and academic structure management, organization governance, document processing, event and attendance tracking, and inventory rental management. Each table stores specific data entities while maintaining structured relationships through primary and foreign key constraints to enforce referential integrity.

The Users table serves as the foundational entity for authentication and role management. It stores user identification details, institutional affiliation, academic program reference, contact information, encrypted credentials, account type, and activity status. Each user is uniquely identified by a primary key. Foreign key relationships link users to Institutes and Academic Programs, ensuring that all accounts are institutionally classified and program-specific.

Academic structure is defined through the Institutes, Academic Programs, and Program Org Mappings tables. Institutes represent academic divisions, while Academic Programs are linked to specific institutes. The Program Org Mappings table connects organizations to academic programs, enabling structured alignment between student organizations and their corresponding academic departments. This modular design ensures scalability and prevents redundancy in academic data classification.

Student identity management is supported through the Student Numbers table, which links student records to programs and institutes. This structure ensures controlled student validation and prevents duplication of student information across organizational and attendance modules.

Organization governance is structured through the Organizations, Organization Members, and Org Roles tables. The Organizations table stores organization profile information, including name, logo, status, and timestamps. The Organization Members table establishes many-to-many relationships between users and organizations, allowing users to hold membership records. The Org Roles table defines role classifications within each organization, enabling role-based access and administrative permissions. Foreign key constraints ensure that membership and role assignments correspond to valid users and organizations.

The Events table manages organization-related activities, storing event details such as name, description, location, schedule, publication status, and

associated organization references. Event creation is linked to authorized users through foreign keys, ensuring accountability. The Attendance Records table records participant time-in and time-out logs, linking attendance to events and users. This relational structure supports accurate attendance tracking and activity validation.

Document workflow management is handled through the Document Submissions, Documents Approved, and Document Annotations tables. The Document Submissions table stores uploaded documents, including metadata such as document type, title, semester, academic year, and submission timestamps. Review and approval processes are tracked using foreign key references to reviewing users. The Documents Approved table stores finalized records of approved submissions, ensuring a separate structured archive. The Document Annotations table supports review transparency by storing feedback comments and structured annotation data linked to specific submissions. This separation of concerns enhances auditability and preserves document lifecycle integrity.

System announcements are managed through the Announcements table, which stores organization-based announcements, publication status, and target audience classification. Foreign key relationships ensure that announcements are associated with valid organizations and creators.

The rental and inventory management module consists of Inventory Categories, Inventory Items, Rentals, and Rental Items tables. Inventory Categories classify rentable assets. Inventory Items store item details, including barcode, cost, category classification, and payment tracking fields. The Rentals table records transaction-level rental data, including rental duration, return timestamps, total cost, payment method, and transaction status. The Rental Items table supports a many-to-many relationship between rentals and inventory items, allowing multiple items per transaction. This normalized design ensures accurate tracking of item availability, payment status, and overtime calculations.

Pending registration processes are managed through the Pending Registrations table, which temporarily stores applicant data awaiting administrative review. This structure separates unverified accounts from the primary Users table, maintaining system integrity and preventing unauthorized access.

The logical database design applies normalization principles up to the Third Normal Form (3NF). Each table stores logically independent data attributes, eliminating redundancy and ensuring that non-key attributes depend solely on the primary key. Many-to-many relationships are resolved through associative tables such as Organization Members and Rental Items, preventing data duplication and maintaining structural clarity.

Primary and foreign key constraints enforce referential integrity across all modules. These constraints prevent orphan records, invalid references, and

inconsistent data entries. Cascading relationships and structured indexing further support efficient query processing and transactional reliability.

Overall, the logical database design provides a scalable, normalized, and relationally consistent foundation for managing student organization governance, digital document workflows, attendance tracking, announcements, and rental services. By structuring data into logically defined tables with enforced relational constraints, the system ensures accuracy, maintainability, performance efficiency, and institutional data governance support.

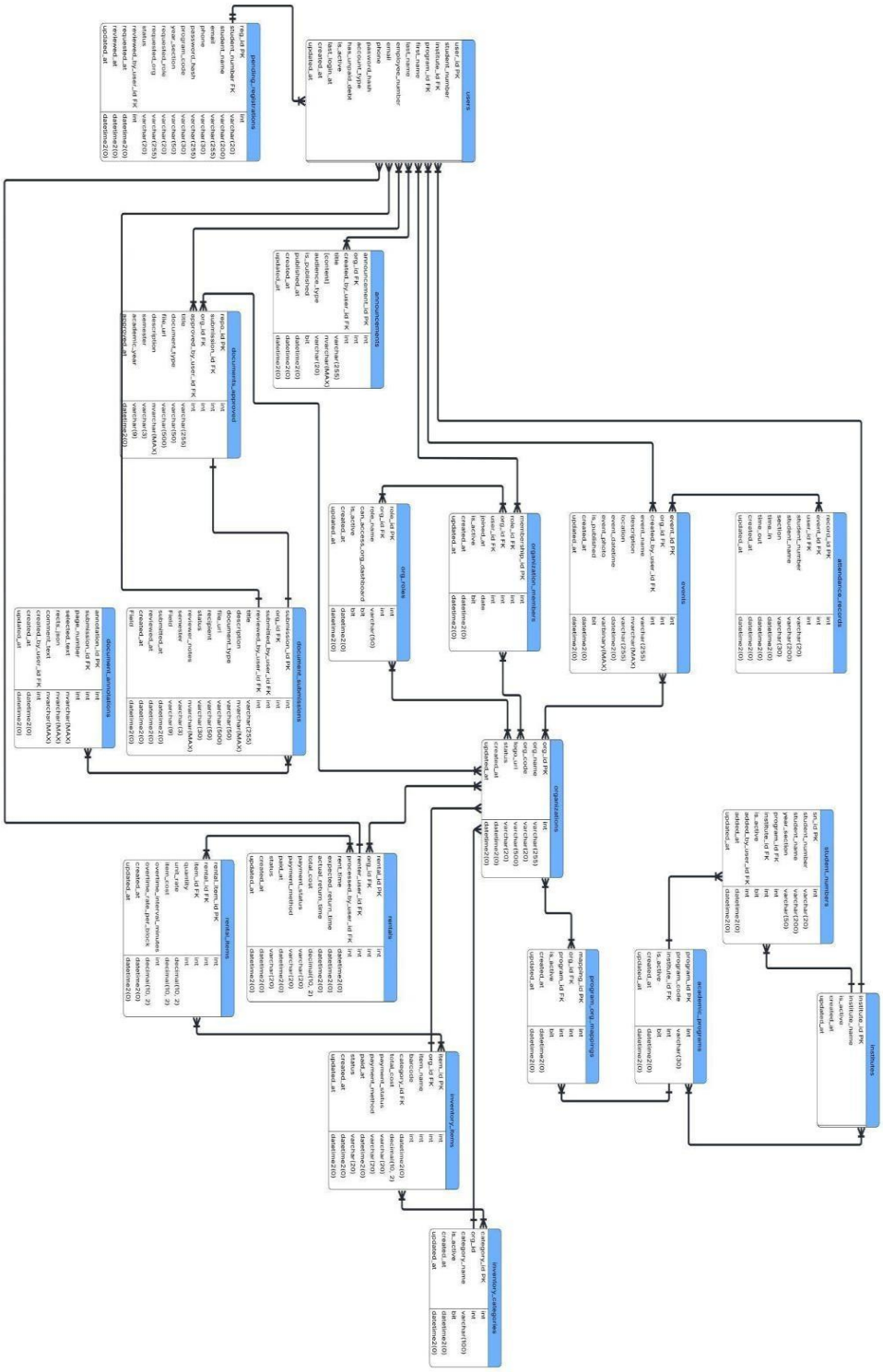

Figure 9. Logical Database Design

System Security and Data Privacy Design

The proposed Integrated Management System for the National Aviation Academy of the Philippines – Villamor (NAAP–Villamor) is designed to centralize student organization management while ensuring strong security controls, data protection, and regulatory compliance. Since the platform handles student information, organizational records, rental transactions, approval workflows, monitoring dashboards, and administrative data, security and privacy mechanisms are embedded directly into the system architecture. The design emphasizes controlled access, encryption, privacy safeguards, and auditability to ensure governance efficiency, accountability, and institutional data protection.

Access Control and Authentication Mechanisms

The system adopts a role-based and identity-centered access control model to ensure that users can only access functions and information relevant to their institutional responsibilities. Users are categorized into defined roles, including students, student organization officers, Office of Student Affairs (OSA) administrators or approvers, and system administrators. Each role is assigned specific permissions based on institutional duties. Students are limited to viewing organization-related information, browsing available services, and performing rental transactions for items offered by student organizations. Student organization officers are authorized to manage their organization's records, manage services offered (such as rentable equipment, materials, or other organization-owned

resources), monitor rental transactions, and upload required organizational reports. OSA administrators are responsible for reviewing and approving organizational reports, monitoring compliance dashboards, and overseeing institutional governance. System administrators are limited to managing technical configurations and maintaining the platform and are not permitted to alter official approval decisions. This separation of duties prevents conflicts of interest, reduces the risk of unauthorized actions, and strengthens accountability within digital workflows.

Authentication mechanisms are implemented to verify user identity before granting access. All users must log in using unique institutional credentials. Passwords are securely stored using non-reversible hashing algorithms such as Argon2id or bcrypt to protect against credential compromise. Multi-Factor Authentication (MFA) is required for privileged roles, including OSA administrators and system administrators, to provide an additional layer of protection against unauthorized access. Secure session management practices are enforced, including HTTP-only and secure cookies, automatic session timeouts after periods of inactivity, and re-authentication for high-risk actions such as approving reports or modifying user roles. These controls reduce the likelihood of unauthorized access and ensure traceable electronic transactions. Account lifecycle management is also implemented, ensuring that user roles are updated or revoked immediately when organizational positions change, thereby maintaining accurate

authorization boundaries.

Data Encryption and Protection Measures

Because the system stores confidential student information, organizational records, rental transaction data, and official reports, a defense-in-depth encryption strategy is applied to protect both data confidentiality and integrity. All communication between users and the web application is secured using HTTPS with TLS 1.3 encryption to prevent interception or tampering during transmission. Legacy protocols are disabled to minimize vulnerabilities associated with outdated encryption standards.

Data stored within the system is protected through encryption at rest. The relational database is encrypted, and uploaded organizational documents are stored in encrypted object storage repositories. Backup files are likewise encrypted using separate encryption keys to ensure that stored data remains protected even in recovery scenarios. Encryption keys are managed through secure key management practices, including restricted administrative access, periodic key rotation, and separation of encryption keys from application code. Database credentials, API keys, and other sensitive configuration data are stored securely and are not embedded directly in source code. These measures significantly reduce the risk of large-scale data exposure in the event of unauthorized access.

To preserve the integrity of governance records, the system prevents unauthorized modification of finalized reports and approved organizational records. Once an approval decision has been recorded, the workflow state cannot be altered without creating a new version entry. Hashing or digital verification techniques may be applied to finalized records to detect tampering. These integrity controls ensure that approval histories, compliance records, and rental monitoring data remain reliable and verifiable for administrative oversight.

Data Privacy and Ethical Considerations

While centralization improves accessibility and monitoring, it also increases privacy risks by consolidating student-related information and rental transaction data into a single digital environment. To address this, the system follows privacy-by-design principles. Data minimization is strictly enforced, meaning only information necessary for student organization management, rental processing, service monitoring, and administrative oversight is collected. Each data element has a clearly defined purpose to prevent unnecessary data accumulation.

Role-scoped visibility ensures that personal and organizational data is accessible only to authorized users. Students cannot access confidential organizational reports or internal records of other organizations. Officers can only manage and view records related to their own organization and its offered services. OSA administrators may access identifiable information strictly for governance,

compliance, and monitoring purposes. Analytics dashboards prioritize aggregated data to minimize exposure of personally identifiable information and reduce privacy risks.

Retention policies are established to prevent indefinite storage of personal data. Rental transaction records and organizational documents are archived or securely deleted after defined institutional retention periods. Procedures are also implemented to address data access, correction, and complaint requests in accordance with applicable privacy regulations. In the event of a security incident, breach response protocols include detection, containment, documentation, and notification procedures. These measures ensure that digital operations enhance institutional efficiency while respecting individual privacy rights.

Audit Logging and Activity Monitoring

To strengthen transparency, traceability, and compliance, the system incorporates comprehensive audit logging and activity monitoring mechanisms. The platform automatically records login attempts, role assignments and modifications, service creation or updates, rental transactions, workflow status transitions, approval actions, and administrative configuration changes. Each log entry includes the user identifier, timestamp, action performed, and reference to the affected record.

Audit logs are stored separately from primary application data and are protected from modification or deletion by standard users. Access to logs is restricted to authorized administrators for compliance verification and incident investigation. Suspicious activities, such as repeated failed login attempts or unauthorized privilege changes, trigger system alerts for review. Logs are encrypted and retained according to institutional retention policies, and secure backups are maintained to ensure availability during recovery scenarios.

Through structured audit logging, encrypted storage, secure authentication, and controlled authorization, the system establishes a secure and accountable digital environment. These security and privacy mechanisms ensure that the Integrated Management System not only improves administrative efficiency and service accessibility but also protects student, rental, and institutional data in accordance with modern security standards and regulatory requirements.

Implementation Environment and Technology Stack

The implementation environment and technology stack form the technical foundation of the integrated student organization management system. In higher education digital platforms, the selection of appropriate software technologies, hardware configuration, architectural design, and deployment infrastructure significantly influences system performance, security, scalability, and maintainability. For a centralized web-based system that supports rental services,

role-based access control, monitoring dashboards, and administrative governance, the chosen environment ensures secure transaction processing, efficient data management, and accessible service delivery across multiple user devices.

Software Requirements

The Integrated Student Organization Management System for NAAP–Villamor operates as a browser-based web application on a PHP/MySQL backend. The software stack is selected to support secure authentication, role-based access, REST API operations, server-rendered pages, and responsive user interfaces.

A web server environment (Apache with PHP) hosts the application, manages client–server communication, and executes business logic for authentication, rentals, approvals, and dashboards. PHP handles server-side rendering for core pages and exposes RESTful endpoints for dynamic interactions.

A relational database management system (MySQL) stores user accounts, role assignments, rental transactions, organizational records, approval states, and audit entries. The database enforces structured schemas, primary/foreign keys, and uses prepared statements for secure data access.

Client-side technologies include HTML5 for structure, CSS for responsive and consistent styling, and JavaScript for interactive behavior and asynchronous

API calls (fetch/AJAX). Modern web browsers (desktop and mobile) are required to render pages, handle forms, and execute client-side scripts securely over HTTPS.

Security controls are supported across the stack: HTTPS/TLS for encrypted transport in production, PHP session management with secure cookies, password hashing (bcrypt/Argon2id), and server-side authorization checks on all API endpoints. Configuration separates environments (development vs. production) so that production instances run with HTTPS, secure cookies, and restricted secrets.

Supporting tools include package/dependency management for client libraries where needed, and backup/monitoring utilities for database and application uptime. This software environment provides a stable, maintainable foundation for centralized organization management, rentals, approvals, and governance workflows.

Hardware Requirements

The deployment of the Integrated Student Organization Management System for NAAP–Villamor requires hardware capable of supporting secure operations, concurrent user access, and reliable processing of organizational records, rentals, and approvals. The hardware baseline prioritizes availability, performance, and data reliability for production use.

The system is hosted on server-class hardware (or an equivalent cloud instance) that runs the Apache/PHP/MySQL stack. The server provides sufficient CPU resources to handle simultaneous sessions from students, organization officers, and OSA administrators, along with real-time dashboard updates and file uploads. Adequate system memory is provisioned to support PHP application execution, database caching, and responsive page/API delivery during peak activity.

Persistent storage uses enterprise-grade or cloud-backed volumes to retain user accounts, role assignments, rental transactions, organizational documents, and approval records. Storage is sized for current usage with headroom for growth and supports fast read/write performance. Regular backups and, where available, redundant storage configurations are employed to protect against data loss and to enable recovery.

Network infrastructure provides stable, low-latency connectivity between client devices and the server environment. Bandwidth is sufficient to accommodate concurrent logins, data submissions, and document uploads, with encrypted transport (HTTPS) for all production traffic. Network controls restrict administrative interfaces to authorized users and protect the application from unauthorized access.

Client devices require only standard hardware capable of running a modern web browser. Desktops, laptops, tablets, and smartphones can access the system

without specialized equipment, benefiting from responsive layouts for varying screen sizes.

By provisioning capable server resources, reliable storage with backups, secure and performant networking, and browser-capable client devices, the hardware environment supports the system's scalability, availability, and secure handling of organizational and student data.

Development Tools and Programming Technologies

The system adopts a monolithic hybrid architecture that combines server-rendered pages with RESTful API endpoints within a unified backend structure. In this configuration, core modules—such as authentication, rental management, service administration, workflow monitoring, and audit logging—operate within a single application codebase while maintaining logical modular separation. Server-rendered pages manage most user interface rendering directly through PHP, while REST APIs handle asynchronous operations, dynamic data updates, and structured communication between the client and server. This hybrid approach supports efficiency, simplified deployment, and cohesive integration between rental services and administrative functions.

Front-end development uses HTML, CSS, and JavaScript to create structured, interactive, and responsive interfaces. JavaScript enhances user experience by enabling dynamic form validation, real-time updates, and

asynchronous data retrieval through REST API calls. The backend, powered by PHP, enforces authentication, manages session control, processes rental transactions, validates inputs, and communicates securely with the MySQL database. The relational database structure ensures normalized storage of user data, rental records, service listings, and monitoring logs. This technology stack balances simplicity, maintainability, scalability, and security, making it appropriate for a centralized academic management platform.

Deployment Environment

During development, the system operates within a local server environment using XAMPP, which integrates Apache, MySQL, and PHP into a unified hosting package. Apache serves as the web server, PHP executes backend logic, and MySQL manages database operations. This environment enables controlled testing, debugging, and validation of authentication processes, rental workflows, and administrative monitoring features prior to production release.

For operational deployment, the system utilizes a cloud-based hosting environment that supports scalability, remote accessibility, and centralized data management. The deployment configuration maintains Apache, PHP, and MySQL components within a secure cloud infrastructure. Cloud hosting supports HTTPS configuration, encrypted communication, automated backups, and consistent uptime reliability. It also enables authorized users to access the system remotely

across desktops, laptops, tablets, and mobile devices. This deployment approach reduces infrastructure maintenance overhead while accommodating growth in rental transactions, organizational data, and user accounts.

Overall, the implementation environment and technology stack consist of HTML, CSS, and JavaScript for the client interface; PHP and MySQL for backend processing and database management; a monolithic hybrid architecture combining server-rendered pages and REST APIs; local development using XAMPP (Apache + MySQL + PHP); and cloud-based deployment for production use. This integrated technological configuration supports centralized rental service management, role-based access control, secure transaction processing, multi-device accessibility, and sustainable digital governance within a higher education institution.